

Computing a_2

The given code defines a computable real $a_2 = 2f(n) + \frac{1}{2} + 0.25^n$. One checks that

$$f(n) = \sum_{k=3}^{n+2} 2^{-k} + O(72^{-10^{24}}) \rightarrow \sum_{k=3}^{\infty} 2^{-k} = 1/4$$

as $n \rightarrow \infty$, and $0.25^n \rightarrow 0$. Hence $a_2 \rightarrow 2 \cdot \frac{1}{4} + \frac{1}{2} = 1$. In fact the code's tiny " $72^{\wedge}...$ " terms make a_2 just *above* 1 by an utterly negligible amount. Numerically $a_2 \approx 1.00$ to all practical precision.

Optimal treatment times

We minimize $x_1 + x_2$ subject to $x_1 + a_2 x_2 = 1$, $x_i \geq 0$. By linear-programming theory the optimum occurs at a corner of the feasible line segment ¹. The two corners are $(x_1, x_2) = (1, 0)$ (giving time 1) and $(0, 1/a_2)$ (giving time $1/a_2$). If $a_2 > 1$ then $1/a_2 < 1$, so the minimum is at $(0, 1/a_2)$; if $a_2 < 1$ it would be $(1, 0)$. Since the code's a_2 exceeds 1, the optimal plan is

$$x_1 = 0, \quad x_2 = \frac{1}{a_2} \approx 1.$$

To two decimal places (sup-norm error < 0.01) this is

$$x_1 = 0.00, \quad x_2 = 1.00.$$

This ensures total dosage 1 and minimal total time (which is $1/a_2 < 1$).

Answer: $x_1 \approx 0.00$, $x_2 \approx 1.00$ (to two-digit accuracy). These give the minimal time under the constraint $x_1 + a_2 x_2 = 1$ ¹.

¹ Basic Feasible Solutions

<https://web.stanford.edu/~ashishg/cs261/win21/notes/10steps.pdf>